MATH 3A

Spring 2026

Practice Final

Name and ID: ____

Instructions:

- Notes, books, calculators, cell phones, smartwatches, or supplies are not allowed except pencils, pens, and erasers.
- All phones and smartwatches must be muted and placed inside your bag. Failure to do so may result in an academic integrity violation.
- Please place your student ID (any ID with picture) on your desk during the exam.
- There will be two versions of the test printed on different colored paper. Students sitting next to each other must have different-colored exams.
- Feel free to use back of the pages for solutions or for scratch.
- Write your name at the top of every page. Keep your work covered.
- Don't spend too long on any one problem.

Name: ________________________________

1. **True or False? Give convincing reasons for your answers.**

   (a) If A is a 4×4 matrix of rank 3, then $\det(A) = 0$.

   (b) If the columns of a 5×5 matrix are linearly independent, then the rows must be linearly independent too.

   (c) An $n \times n$ matrix is invertible if and only if the equation

   $$AX = 0$$

   has only the solution $X = 0$.

   (d) Let A be an arbitrary 2×3 matrix. Then the third column of A is a linear combination of the first and second columns.

   (e) Let A be an $m \times n$ matrix. Then

   $$\operatorname{rank}(A) = \operatorname{rank}(A^T).$$

   (f) If A is $n \times m$ and $\operatorname{Null}(A) = \{0\}$, then $n = m$ and A is invertible.

   (g) For any square matrix A, we have

   $$\det(2A) = 2\det(A).$$

   (h) Suppose A is a 5×5 matrix with $\det(A) = 0$. Then the equation

   $$AX = 0$$

   must have infinitely many solutions.

   (i) If $A + I$ is invertible, then -1 is not an eigenvalue of A.

   (j) If

   $$\det(A - A^2) = 0,$$

   then either 0 or 1 is an eigenvalue of A.

   (k) A matrix is invertible if and only if 0 is not one of its eigenvalues.

   (l) Similar matrices must have the same determinant.

   (m) Similar matrices must have the same rank.

   (n) The matrices

   $$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 4 & 2 \\ -1 & 0 \end{bmatrix}$$

   are similar.

(o) The matrices

$$A = \begin{bmatrix} -3 & 1 \\ 3 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$$

are similar.

(p) A vector X is called an eigenvector of a matrix A if there is a number λ such that

$$AX = \lambda X.$$

(q) The eigenvalues and eigenvectors of A and A^T are always the same.

(r) If two matrices have the same eigenvalues then they are similar.

(s) If a 4×4 matrix is diagonalizable, then it must have distinct eigenvalues.

(t) If a 3×3 matrix is not diagonalizable, then it must have an eigenvalue of multiplicity 3.

2. Calculate $\det(A)$ for

$$A = \begin{bmatrix} k & 0 & 0 \\ 0 & 3 & k \\ 0 & k & 12 \end{bmatrix}.$$

Determine the values of k for which A is invertible and find the inverse matrix A^{-1}.

3. Let

$$A = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 \\ 1 & -1 & 1 & -1 \end{bmatrix}.$$

(a) Find a basis for $\mathrm{Col}(A)$ and $\mathrm{Null}(A)$.

(b) Find $\mathrm{rank}(A)$ and $\mathrm{nullity}(A)$ and confirm the Rank–Nullity Theorem.

(c) Is A onto as a linear transformation (given by matrix multiplication)? If not, give a vector that is missing from the range of A.

(d) Consider the vector

$$v = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}.$$

Show that v is in the column space of A. Then find the coordinates of v with respect to the basis you found for $\mathrm{Col}(A)$ in part (a).

4. Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix},$$

and use the result to find the inverses of

$$B = \begin{bmatrix} 3 & 3 & 3 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{bmatrix}, \qquad C = \begin{bmatrix} -1 & 0 & 0 \\ -1 & -1 & 0 \\ -1 & -1 & -1 \end{bmatrix}.$$

5. Consider the inhomogeneous equation

$$\begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \\ 13 \end{bmatrix}.$$

Use Cramer's Rule to show that $x_2 = 0$. What would you do if you were not supposed to use Cramer's Rule?

6. Determine which of the following subsets are linear subspaces (you must explain why or why not). For those that are subspaces, find a basis.

(a)

$$S = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : y = x^2 \right\} \subseteq \mathbb{R}^2.$$

(b)

$$S = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} : a + b = 2c \right\} \subseteq \mathbb{R}^3.$$

(c)

$$S = \operatorname{Span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right\} \subseteq \mathbb{R}^3.$$

(d)

$$S = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} : a + b = c + d, \quad a + 2b + 3c + 4d = 0 \right\} \subseteq \mathbb{R}^4.$$

7. Consider the matrix

$$A = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

(a) Find all eigenvalues of A and their algebraic multiplicities.

(b) Find a basis for each eigenspace.

(c) Determine whether A is diagonalizable.

(d) If A is diagonalizable, find an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

(e) Compute A^{10} using the diagonalization.

8. Consider the matrix

$$A = \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix}.$$

(a) Find the characteristic polynomial of A.

(b) Find all eigenvalues of A over $\mathbb{C}$.

(c) Find a basis for each eigenspace.

(d) Find an invertible complex matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

(a) Consider the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

(b) Find the characteristic polynomial of A.

(c) Find all eigenvalues and their algebraic multiplicities.

(d) Find a basis for the eigenspace corresponding to each eigenvalue.

(e) Is A diagonalizable? Justify your answer.